JL COLLECTION Premium Treatment Oil Formulation & Competitive Discussion

Evidence-based review of the proposed botanical formula, lightweight sensory target, competitive pricing, treatment positioning, weaknesses and validation priorities.

Important: the current list contains ingredient names only - no percentages, extract grades, supplier specifications or finished-formula data. Conclusions on feel, safety and efficacy are therefore directional, not a final formulation verdict.

DECISION SUMMARY

Executive conclusion

What is strong

A credible South Asian botanical scalp-treatment story: rosemary + amla + coconut, with familiar traditional ingredients and clear pre-wash ritual potential.

What is weak

As written, the system is likely to be richer/heavier than premium “weightless oils,” because the carrier list is dominated by conventional triglyceride oils and includes several botanicals with weak direct hair evidence.

What I would change

Keep the botanical identity, but simplify the active story and build it on a lighter carrier architecture. Thuja is the clearest removal candidate; mustard and arnica need strong safety justification; shikakai/reetha fit rinse-off products better.

Recommended product architecture

Priority	Recommended direction	Reason
1	Lightweight botanical treatment oil - not a “pure traditional oil blend”	Matches the user requirement and closes the sensory gap vs Mielle Light/K18/Olaplex.
2	Rosemary + amla + coconut as the core evidence/story spine	Rosemary has limited direct human evidence; coconut has useful hair-fiber evidence; amla is promising but mostly indirect for topical oil.
3	Re-evaluate Thuja, mustard, arnica; move shikakai/reetha to rinse-off if needed	Safety/evidence/formulation-fit concerns are larger than the incremental marketing value.
4	Do not lead with “hair regrowth” without product-specific clinical data	The current evidence does not justify a strong medical-style growth claim for the finished blend.

Sources: [12], [13], [16], [17], [22] | Judgment: evidence quality varies by ingredient and by endpoint.

Proposed JL COLLECTION formula: what is actually defined

- Arnica montana
- Rosemary oil
- Jaborandi (Pilocarpus spp.)
- Shikakai (Acacia concinna)
- Amla (Phyllanthus emblica)
- Reetha / soapnut (Sapindus mukorossi)
- Mustard oil
- Olive oil
- Sweet almond oil
- Coconut oil
- Thuja (species/extract not specified)

Critical missing information before formulation approval

- Exact percentage of every ingredient and carrier.
- Botanical form: essential oil, infused oil, CO2 extract, tincture, powder, standardized extract, etc.
- Supplier COA/specifications, allergens, pesticides/heavy metals, oxidation/peroxide values.
- Confirmed species and thujone profile for “Thuja.”
- Whether “100% natural / silicone-free” is a hard brand requirement.
- Intended use: pre-wash scalp oil, leave-in length oil, overnight treatment, or one product for all three.

Current research assumption

“Thuja” is treated as Thuja occidentalis for risk screening only. This must be confirmed.

No percentages = no reliable final prediction of greasiness, irritation or stability.

MARKET BENCHMARK

Competitive reference set

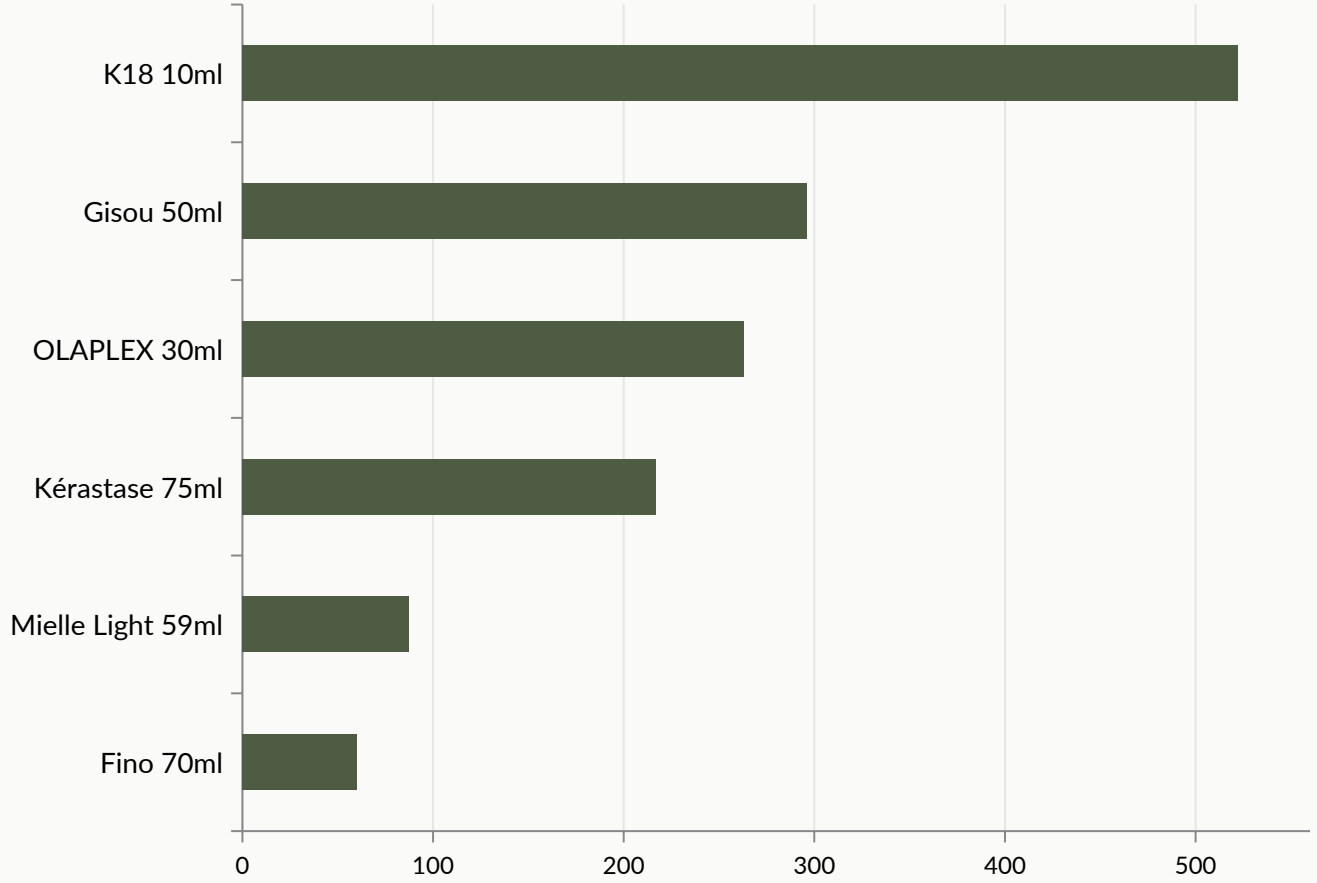
Product	Role / treatment promise	Leading architecture	Scalp use?	Observed PK price
Mielle Rosemary Mint Light 59ml	Scalp + strengthening + lightweight moisture	Grapeseed, soybean, almond, castor, jojoba + rosemary/mint	Yes	Rs 5,160
OLAPLEX No.7 30ml	Bond repair appearance + shine + frizz + heat	Dimethicone + isohexadecane + isoalkane + ester	Mostly lengths	Rs 7,900
K18 Molecular Repair Oil 10ml	Damage repair + frizz + shine + heat	Light ester + squalane + avocado/sunflower + alkanes	Lengths	Rs 5,220
Kérastase Elixir Ultime 75ml	Luxury shine + frizz + heat + strengthening	Isododecane + silicones + camellia/argan/marula oils	No	Rs 16,250
Gisou Honey Infused 50ml	Hydration + shine + frizz / luxury	Silicones + honey + almond/coconut/argan oils	Lengths	Rs 14,800
Fino Premium Touch 70ml	Repair appearance + shine + frizz / finishing	Mineral oil + silicone + isododecane + squalane	Lengths	Rs 4,199

Interpretation: Mielle Light is the closest benchmark for a botanical scalp oil. OLAPLEX/K18/Kérastase are the better benchmarks for “light, non-greasy” sensory engineering.

Sources: [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11] | Prices are observed retail snapshots; promotions, stock and authenticity channels can change.

COMMERCIAL BENCHMARK

Pakistan price comparison



What the chart means

Premium imported oils sell far above the ingredient-cost logic of traditional oil. Their pricing is supported by sensory performance, claims, brand equity, testing and packaging.

JL price is not yet defensible

Without final COGS, pack size, stability data, sensory results and product proof, an exact launch price would be guesswork.

Working discussion range only

For a 100ml premium local treatment, Rs 3,499-4,499 would equal ~Rs 35-45/ml - below these imported benchmarks. Validate willingness-to-pay before fixing.

Sources: [2], [4], [6], [8], [10], [11] | Calculated from observed shelf prices; K18 is a 10ml mini and therefore has a high per-ml figure.

FORMULATION COMPARISON

Ingredient architecture: JL vs lightweight leaders

Formula	What is doing most of the sensory work	Botanical story	Likely feel / role
JL concept as listed	Mustard + olive + almond + coconut (percentages unknown)	Very high: 11 named botanicals/oils	Risk of rich/oily feel if conventional oils dominate
Mielle Light	Grapeseed + soybean + almond; castor lower; jojoba	High: rosemary, mint + many oils/extracts	Closest natural-oil benchmark for scalp + daily use
OLAPLEX No.7	Dimethicone + isohexadecane + C13-14 isoalkane + coco-caprylate	Low/moderate	Fast spread, polished slip, high shine; styling-first
K18 Oil	Caprylyl caprylate/caprate + squalane + C13-15 alkane/undecane/tridecane	Moderate	Silicone-free but engineered for very light spread
Kérastase Elixir	Isododecane + dimethicone/dimethiconol, then multiple oils	Moderate	Volatile/light carrier + silicone finish; luxury lengths oil
Gisou Honey Oil	Dimethicone + dimethiconol + trisiloxane before honey/oils	Moderate/high storytelling	Richer than K18/Olaplex; still silicone-engineered

Core difference: competitors do not rely on “many oils” to create premium performance. They engineer the carrier system first, then layer the hero ingredients into it.

Sources: [1], [3], [5], [7], [9] | INCI order is qualitative, not a disclosed percentage formula.

Why the premium oils can feel lighter

“Light” is primarily a formulation-architecture problem, not a marketing adjective.

Conventional triglyceride oils

Mustard, olive, almond, coconut

Good emolliency and heritage value, but a blend dominated by these can leave more persistent oily residue, especially on fine/low-porosity hair.

JL relevance: high risk

Light botanical carriers

Grapeseed, sunflower, jojoba-type choices

Mielle Light starts with grapeseed and soybean rather than a heavy castor/coconut base. It remains a botanical oil, but the carrier hierarchy is different.

JL relevance: high

Modern light emollients

Squalane, coco-caprylate/caprate, light esters, C13-15 alkane / hemisqualane

K18 demonstrates a silicone-free route to a highly spreadable, low-residue feel.

JL relevance: high

Volatile + silicone systems

Isohexadecane / isododecane + dimethicone / dimethiconol

OLAPLEX and Kérastase use this route for slip, shine, heat styling and a less greasy finish.

JL relevance: optional

Best fit for JL: botanical identity + modern light emollients. This can remain silicone-free while getting closer to the requested “lighter, not greasy” feel.

Sources: [1], [3], [5], [7] | Sensory prediction remains provisional until percentages and prototypes are tested.

EVIDENCE QUALITY

Evidence scorecard for the proposed ingredients

Ingredient	Best-supported role	Direct hair-growth evidence	Main concern / limitation
Rosemary oil	Scalp / possible AGA support	LIMITED HUMAN	One 6-month randomized comparison vs 2% minoxidil; no placebo and not evidence for every cause of hair fall.
Coconut oil	Hair-fiber protein-loss protection	NO DIRECT GROWTH	Good fiber evidence; can feel heavy if overused.
Amla	Traditional tonic / antioxidant story	INDIRECT	Oral RCT and preclinical/topical-delivery work; not strong proof for amla oil itself.
Olive oil	Emollient / conditioning	NONE FOUND	Repeated topical olive oil impaired adult forearm barrier in one trial; scalp relevance is indirect.
Sweet almond oil	Emollient / slip	NONE FOUND	Useful carrier, not a demonstrated growth active.
Mustard oil	Traditional warming oil	NONE CONVINCING	Allyl isothiocyanate is a human skin irritant; odor/sensory burden.
Arnica montana	Traditional topical botanical	NONE CONVINCING	Documented contact sensitization risk; weak hair-specific evidence.
Jaborandi	Traditional hair tonic	VERY LIMITED	Pilocarpus is a pilocarpine source; a cosmetic extract is not equivalent to a clinically dosed drug.
Shikakai	Cleansing / saponin source	NONE	Stronger logic in rinse-off shampoo than leave-on anhydrous oil.
Reetha	Cleansing / saponin source	NONE	Same formulation-fit issue as shikakai.
Thuja*	No well-supported hair role	NONE CONVINCING	Thujone-related toxicology; species/extract/thujone content must be defined.

Sources: [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22] | *Assuming Thuja occidentalis for screening; verify supplier identity. “None found” = no convincing direct human evidence identified in this review.

Where JL is genuinely differentiated - and where it is not

Keep as core

ROSEMARY
Best direct human relevance in the set, but still limited.

COCONUT
Best hair-fiber damage evidence in the set.

AMLA
Strong heritage and some emerging evidence, but topical claim should remain conservative.

Useful mainly as carriers / story

ALMOND + OLIVE
Primarily emollient value, not treatment proof. Olive should not dominate scalp leave-on without tolerance testing.

JABORANDI
Differentiating heritage ingredient, but supplier standardization and claim discipline are essential.

Reconsider before prototype 1

THUJA
Weak hair rationale + avoidable toxicology uncertainty.

MUSTARD
Irritation/odor/sensory trade-off.

ARNICA
Sensitization concern + little hair-specific proof.

Move to a companion rinse-off product?

SHIKAKAI + REETHA are chemically rich in saponins and have a clearer cleansing/surfactant role. In an oil, especially a clear lightweight leave-on product, they may add formulation complexity without a strong performance payoff.

Ingredient count is not a quality metric

Eleven hero names can make the story look “traditional blend” rather than modern premium treatment. A smaller, more defensible active system is easier to standardize, test, explain and claim.

Rational direction: fewer hero actives + better carrier engineering + better evidence beats “more herbs.”

Sources: [12], [13], [15], [16], [17], [20], [21], [22]

Current weaknesses and flaws

Issue	Impact	Severity	What to do
No percentages / extract forms	Cannot judge efficacy, greasiness, irritation, stability or COGS reliably	HIGH	Freeze claims and positioning until prototype composition is defined.
Lightweight target conflicts with current carrier list	Mustard/olive/almond/coconut can create a persistent oily film if they dominate	HIGH	Use a lighter carrier backbone; keep richer oils as supporting ingredients.
Thuja is under-defined	Species and thujone profile unknown; no strong hair rationale	HIGH	Remove from first prototype unless toxicologist/supplier data justify it.
Mustard / arnica risk	Potential irritation or sensitization can undermine a leave-on scalp claim	HIGH	Minimize/remove or validate with safety testing and clear supplier specs.
Shikakai + reetha role mismatch	Saponin botanicals are better suited to cleansing/rinse-off systems	MEDIUM	Move to shampoo/cleansing SKU or use only a validated oil-compatible extract.
Weak proof for “hair fall / growth” as a finished blend	Most ingredients have traditional or indirect evidence	HIGH	Use cosmetic claims until product-specific clinical data exists.
No heat/frizz technology	Competitors deliver immediate visible styling benefits, not only long-term treatment story	MEDIUM	Decide whether JL should add measurable anti-frizz/shine/heat-protection performance.

Sources: [12], [15], [16], [17], [20], [21], [22] | Severity is a formulation/business judgment, not a regulatory determination.

PROTOTYPE STRATEGY

Three realistic formulation routes

A. 100% botanical oil

Architecture

Dominant lighter plant oils such as grapeseed/sunflower-type carriers; small supporting amounts of coconut/almond/olive; simplified botanicals.

Trade-off

Strongest “natural oil” story. Can be lighter than traditional oils, but may not match K18/OLAPLEX weightlessness.

Good if “100% botanical” is non-negotiable.

B. Botanical + modern light emollients

Architecture

Light ester / squalane / plant-derived alkane type backbone + targeted botanical oils/extracts. Silicone-free is possible.

Trade-off

Best balance of botanical identity, spreadability, dry-down and premium sensory.

RECOMMENDED for JL.

C. Silicone serum-oil hybrid

Architecture

Volatile hydrocarbon + silicone system with smaller botanical fraction.

Trade-off

Closest route to OLAPLEX/Kérastase/Fino slip, gloss and anti-frizz performance; weakest “pure oil” perception.

Use only if performance outranks natural positioning.

Do not set exact percentages in the meeting. First choose the architecture and claim target; then a cosmetic chemist can optimize the ratios through prototype testing.

Sources: [1], [3], [5], [7]

Treatment positioning comparison

Product	Scalp / roots	Growth / retention story	Damage / breakage	Frizz / shine	Heat styling	Lightweight design
JL concept	Strong	Potential - mostly rosemary story	Some fiber protection via coconut	Unproven in final formula	Not defined	AT RISK
Mielle Light	Strong	Strong marketing; rosemary/biotin	Moderate	Moderate	Not core	Strong
OLAPLEX No.7	Low	None	Strong brand repair claim	Strong	Strong	Strong
K18 Oil	Low	None	Strong brand repair claim	Strong	Strong	Strong
Kérastase Elixir	No scalp	None	Strengthening / cosmetic repair	Strong	Strong	Strong
Gisou	Low	None	Cosmetic nourishment	Strong	Strong brand claim	Medium/richer

JL advantage

Scalp-first botanical treatment is a real position the styling oils do not own.

JL gap

Immediate sensory proof - lightweight, frizz, shine, heat and residue - is not yet designed or validated.

Sources: [1], [3], [5], [7], [9], [12], [13] | Competitor “strong” entries reflect brand product claims unless otherwise stated.

Claims: what is defensible now vs what needs proof

Lower-risk cosmetic claims to test and support

- Lightweight, non-greasy feel
- Nourishes and conditions the scalp
- Helps reduce dryness and roughness
- Helps reduce hair breakage / improves manageability
- Leaves hair softer, smoother and shinier
- Pre-wash treatment that helps protect hair during washing (if coconut-rich performance is confirmed)
- Suitable for regular scalp-care ritual (after safety validation)

Claims that require stronger product-specific evidence

- “Stops hair fall”
- “Regrows hair” / “new hair growth”
- “Treats androgenetic alopecia / alopecia”
- “Clinically proven” without a clinical protocol on the final product
- “Repairs bonds” without validated technology
- Heat protection to a stated temperature without instrument testing
- “Anti-dandruff treatment” without appropriate efficacy / regulatory support

Pakistan has a General Cosmetics Act (2023) covering cosmetic quality, labeling, packing, manufacturing, storage, distribution and sales. Final claims should be reviewed against current local requirements before launch.

Sources: [12], [13], [23]

Recommended usage architecture

Mode 1 - Pre-wash scalp treatment

Apply a controlled amount to scalp and roots; massage; leave for a defined short treatment window; shampoo out.

Why: best fit for the herbal scalp story and coconut's hair-fiber evidence. Also reduces the risk that a richer formula feels greasy all day.

Mode 2 - Targeted length treatment

Use only a few drops on mid-lengths and ends where needed.

Why: allows the "lightweight" promise to be experienced even if the product contains meaningful botanical oil content. Dose control is part of sensory performance.

Avoid one-message-does-everything

A heavy overnight scalp oil and a weightless finishing oil are different formulation jobs. Trying to make one product serve every use can create compromises in feel, claims and dosing.

Best meeting question: Is JL primarily a PRE-WASH SCALP TREATMENT that also feels unusually light - or a DAILY LEAVE-IN OIL with a scalp story? The answer changes the formula.

Sources: [1], [5], [7], [13]

Validation plan before commercial lock

Gate	Minimum work	What it answers
1. Raw-material definition	Supplier COAs; botanical species; extract type; allergen profile; peroxide/oxidation specs	Do we know what is actually in the bottle?
2. Prototype sensory test	At least 3 carrier prototypes benchmarked blind against Mielle Light + one styling oil; score grease at 1/10/30 min, residue, spread, washout	Is “lightweight” true, not just claimed?
3. Stability	Accelerated temperature, freeze-thaw where relevant, light exposure, odor/color, precipitation, packaging compatibility	Will it remain uniform, stable and acceptable through shelf life?
4. Safety / tolerance	Professional safety assessment; controlled patch/compatibility testing; extra attention to essential oils, mustard, arnica, Thuja	Can it be used repeatedly on scalp with acceptable tolerance?
5. Performance	Instrument/consumer tests for shine, combing, breakage, frizz or hair-fall endpoint - choose only the claims you want to make	Which claims can be backed by data?
6. Pilot use test	4-8 week structured consumer test with target hair types; collect adverse events and usage behavior	Does the real-world ritual work in Pakistan climate and routines?

Exact study design should be set by the cosmetic chemist / safety assessor / clinical testing partner.

Recommended direction for prototype 1

Keep the identity

CORE STORY
Rosemary + amla + coconut

SUPPORTING BOTANICALS
Jaborandi only if standardized and strategically valuable; almond as a sensory/emollient support.

OPTIONAL
Olive only at a level justified by feel and scalp tolerance.

Change the architecture

REMOVE / PAUSE
Thuja for prototype 1.

REASSESS
Mustard and arnica; retain only if safety + story payoff justify them.

MOVE OUT OF OIL
Shikakai + reetha to a future cleansing/rinse-off product unless there is a validated oil-soluble extract reason.

ADD
A modern light-emollient backbone if the brand allows it.

Recommendation: Route B - a silicone-free, botanical + modern-light-emollient treatment. It best reconciles the premium "treatment oil" story with the required non-greasy sensory target.

Sources: [1], [5], [12], [13], [15], [16], [17], [20], [21], [22]

DISCUSSION AGENDA

Decisions to make in the physical meeting

1	**Primary job**	Pre-wash scalp treatment vs daily leave-in vs dual use?
2	**Natural standard**	Must it be 100% botanical, or can we use modern plant-derived emollients?
3	**Ingredient discipline**	Agree to pause Thuja; decide mustard/arnica; decide whether shikakai/reetha move to shampoo.
4	**Hero claim**	Scalp nourishment + reduced breakage, or a stronger growth claim that requires clinical investment?
5	**Pack / price**	60ml, 80ml or 100ml? Do not lock price until COGS and dose-per-use are known.
6	**Benchmark**	Approve Mielle Light as scalp/light benchmark and K18/OLAPLEX as sensory-performance benchmarks.
7	**Evidence budget**	What testing budget is available for safety, sensory, stability and claims?

Goal: leave the meeting with an architecture brief, not a marketing slogan.

Sources

[1]	**Mielle Organics**	Rosemary Mint Light Scalp & Hair Strengthening Oil - product, use and INCI	https://mielleorganics.com/collections/daily-care/products/rosemary-mint-light-scalp-hair-strengthening-oil
[2]	**TheSkinFit Pakistan**	Mielle Rosemary Mint Light Oil 59ml - observed Pakistan retail price	https://theskinfit.com/products/mielle-rosemary-mint-light-scalp-hair-strengthening-oil-59ml
[3]	**OLAPLEX**	No.7 Bonding Oil - current product claims and INCI	https://olaplex.com/products/olaplex-n-7-bonding-oil-30ml
[4]	**TheSkinFit Pakistan**	OLAPLEX No.7 Bonding Oil 30ml - observed Pakistan retail price	https://theskinfit.com/products/olaplex-no7-bonding-oil
[5]	**K18 Hair**	Molecular Repair Hair Oil - product, use and INCI	https://www.k18hair.com/products/molecular-repair-hair-oil
[6]	**Aesthetic Life Pakistan**	K18 Molecular Repair Hair Oil 10ml - observed Pakistan retail price	https://aestheticlife.com/products/k18-molecular-repair-hair-oil-30ml
[7]	**Kérastase**	Elixir Ultime Hair Oil - product, claims and INCI	https://www.kerastase-usa.com/collections/elixir-ultime/3474637215170.html
[8]	**Vegas Pakistan**	Kérastase Elixir Ultime Originale 75ml - observed Pakistan price / sold count	https://www.vegas.pk/public/index.php/pd/kerastase-pakistan/kerastase-elixir-ultime-originale-hair-oil-75ml
[9]	**Gisou**	Honey Infused Hair Oil - INCI reference	https://gisou.com/products/honey-infused-hair-oil
[10]	**The Vault Pakistan**	Gisou Honey Infused Hair Oil 50ml - observed Pakistan retail price	https://thevault.pk/products/gisou-honey-infused-hair-oil-50ml
[11]	**Fino Pakistan**	Shiseido Fino Premium Touch Hair Oil 70ml - price, description and ingredient categories	https://www.fino.com.pk/products/shiseido-fino-premium-touch-hair-oil-70ml

Sources

[12]	**Panahi et al., SKINmed 2015**	Rosemary oil vs 2% minoxidil in androgenetic alopecia; randomized comparative trial	https://pubmed.ncbi.nlm.nih.gov/25842469/
[13]	**Rele & Mohile, J Cosmet Sci 2003**	Coconut oil reduced hair protein loss in damaged and undamaged tresses	https://pubmed.ncbi.nlm.nih.gov/12715094/
[14]	**Ruetsch et al., J Cosmet Sci 2001**	Coconut oil penetration into hair fibers	https://pubmed.ncbi.nlm.nih.gov/11413497/
[15]	**Danby et al., Pediatr Dermatol 2013**	Olive vs sunflower oil on adult skin barrier	https://pubmed.ncbi.nlm.nih.gov/22995032/
[16]	**Andersen et al., Pain 2017**	Topical allyl isothiocyanate (mustard oil component) as human irritant model	https://pubmed.ncbi.nlm.nih.gov/28614189/
[17]	**Reider et al., Contact Dermatitis 2001**	Contact sensitization to Arnica montana	https://pubmed.ncbi.nlm.nih.gov/11722485/
[18]	**Akhbari et al., J Ethnopharmacol 2024**	Oral amla RCT in female androgenetic alopecia - indirect for topical oil	https://pubmed.ncbi.nlm.nih.gov/37487962/
[19]	**Chaiyana et al., 2022**	Amla extract transfersomes / follicle-targeting preclinical work	https://pubmed.ncbi.nlm.nih.gov/35781257/
[20]	**Yoshikawa et al., 1997**	Acacia concinna saponins - chemistry supports cleansing/surfactant role	https://pubmed.ncbi.nlm.nih.gov/9145500/
[21]	**Chen et al., 2020**	Sapindus mukorossi saponin fraction - cosmetic additive research	https://pubmed.ncbi.nlm.nih.gov/33152431/
[22]	**Radulović et al., Food Chem Toxicol 2017**	Thujones and thujone-containing essential oils including Thuja occidentalis - toxicology	https://pubmed.ncbi.nlm.nih.gov/28472675/
[23]	**Pakistan Code / Ministry of Law & Justice**	Pakistan General Cosmetics Act, 2023	https://pakistancode.gov.pk/english/UY2FqaJw1-apaUY2Fqa-apaUY2Npa5pnbA%3D%3D-sg-jjjjjjjjjjjj

Evidence review intentionally separates brand claims from independent published evidence.